Segmenting Tumor Images to Study Immune Cells in Endometrial Cancer

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01

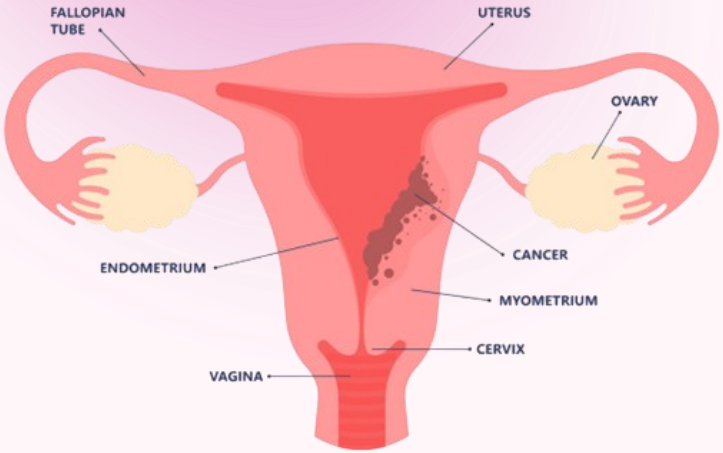
Introduction & Background

Clinical Context

- **Growing Health Challenge**
  Most common gynecological cancer in developed nations

- **Treatment Limitation**
  Advanced stages face poor long-term survival with chemotherapy

- **Immunotherapy Potential**
  High immunogenicity in specific subtypes makes them susceptible to Immune Checkpoint Inhibitors (ICIs)



https://www.vejthani.com/wp-content/uploads/2022/06/Endometrial-cancer-1.jpg

Molecular Subtypes

Biomarker	Profile	Behavior	Mutation Type	Mutation Count
MSI-H	Unstable	"Hot" (Strong Immune Response)	dMMR Mismatch Repair deficiency causes instability	High 10 – 100 mut/Mb
MSS/TMB-L	Stable	"Cold" Poor Response	P53 Structural Changes rather than specific mutations	Low < 10 mut/Mb
MSS/TMB-H	Stable	High Neoantigens	POLE Mutation Broken DNA "proofreading" leads to errors	Ultra High > 100 mut/Mb

Scope of the Project

- Analyze immune cells within the MSS/TMB-H subtype
- Quantify densities and identify unique signatures
- Generate high-resolution Heatmaps of immune cell Distribution
- Develop a Machine Learning model to classify molecular subtypes

02

Methods

Methods

Dataset

HoVer-NeXt

Heatmap & CSV

Exploratory & Statistical Analysis

Machine Learning

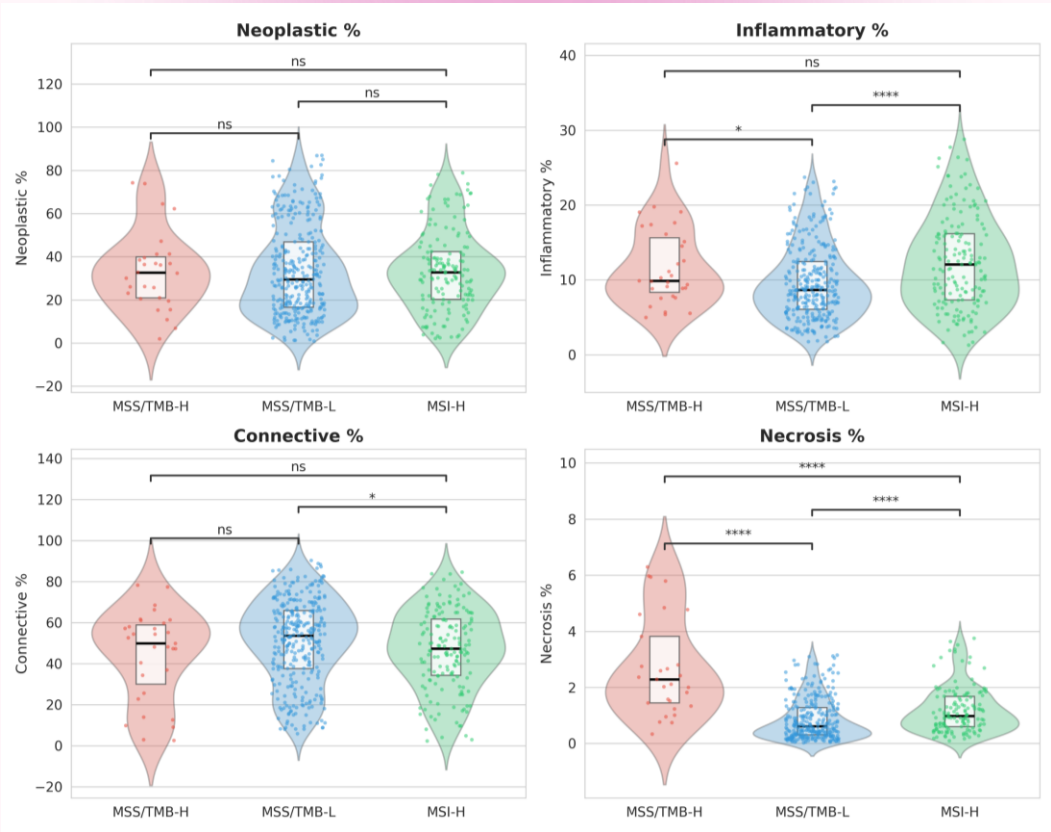
UMAP

03

Results

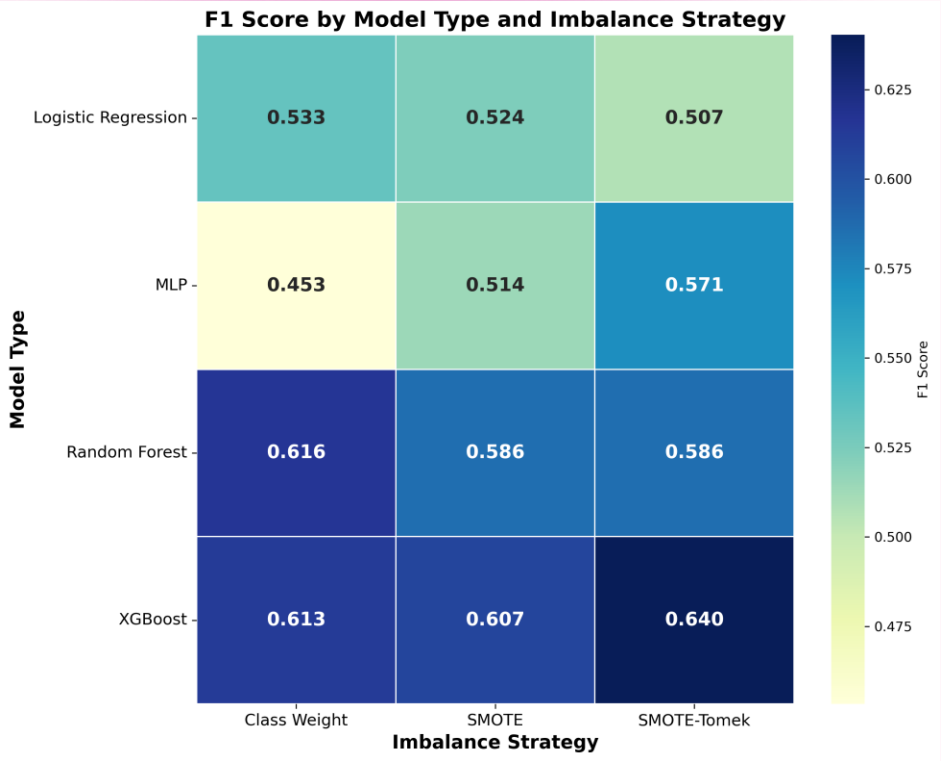
Violin Plots

- Differences between subtypes were confirmed using Kruskal-Wallis tests *($p < 0.05$)*
- Inflammatory percentages alone does not significantly differentiate MSS/TMB-H
- Necrosis emerged as the strongest differentiator with MSS/TMB-H displaying singificant elevated levels

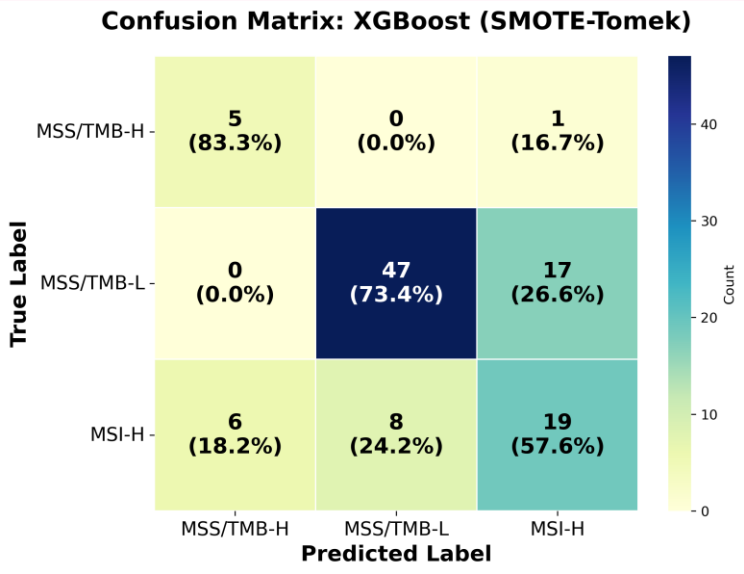


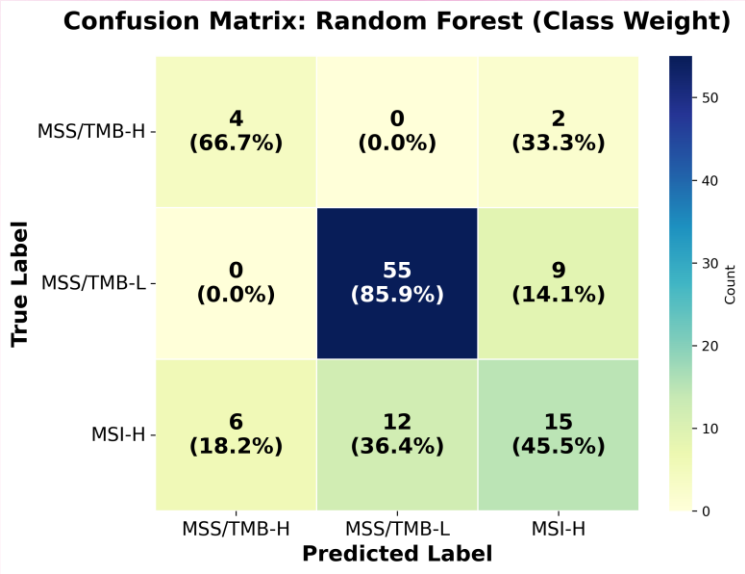
Machine Learning

- 12 Models were Trained
- XGBoost with SMOTE-Tomek achieved the best results (F1 = 0.64)
- Random Forest with class weight was the second best (F1 = 0.62)



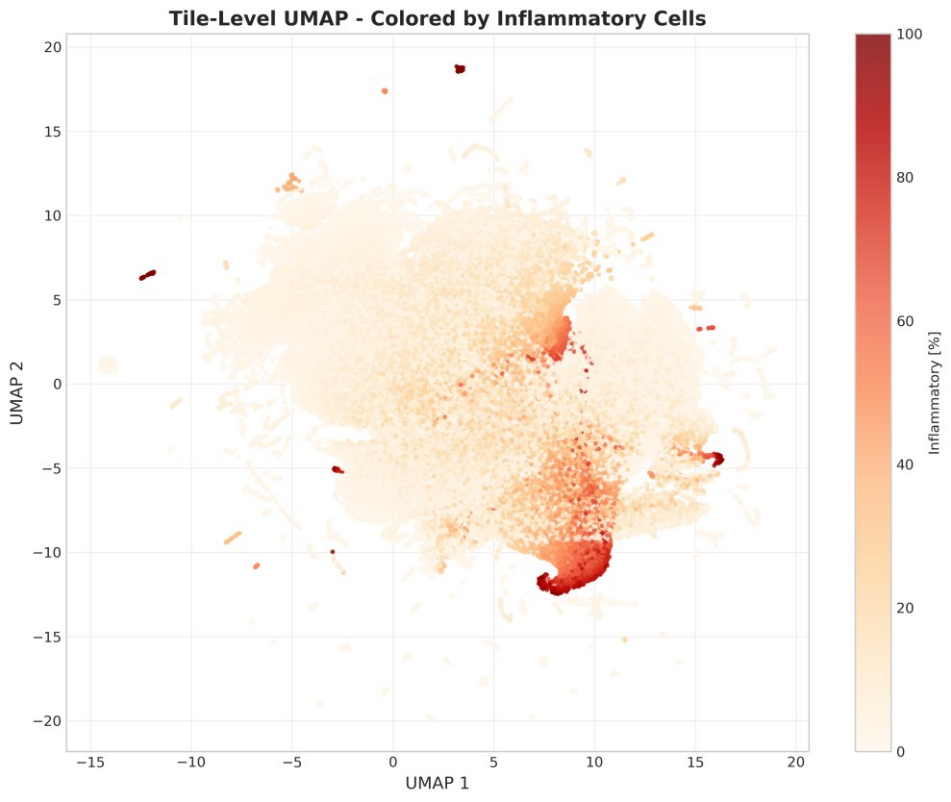
Confusion Matrices





UMAP

- HoVer-NeXt successfully distinguishes immune cells
- Two prominent clusters suggests specific microenvironmental niches
- RGB composite mapping to visualize interactions



04

Discussion

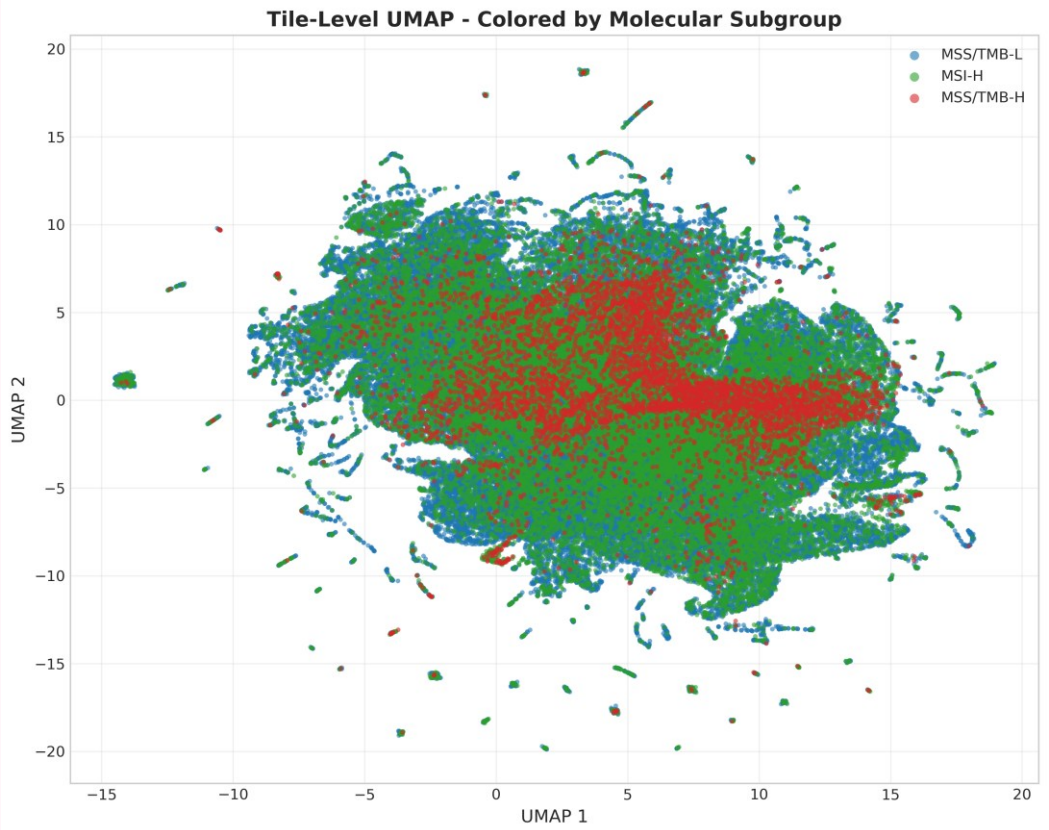
Discussion

- **Necrosis** emerged as the strongest statistical differentiator, significantly elevated in MSS/TMB-H
- Machine learning models successfully predicted molecular subtypes with good accuracy
- Inflammatory **counts alone are insufficient,** spatial distribution is required to distinguish this subtype
- Confusion matrices revealed shared immunogenic features between MSS/TMB-H and MSI-H

05

Questions

UMAP


Tile-Level UMAP - Colored by Molecular Subgroup
MSS/TMB-L
MSI-H
MSS/TMB-H
UMAP 2
UMAP 1
20
15
10
5
0
−5
−10
−15
−20
−15
−10
−5
0
5
10
15
20

Sources

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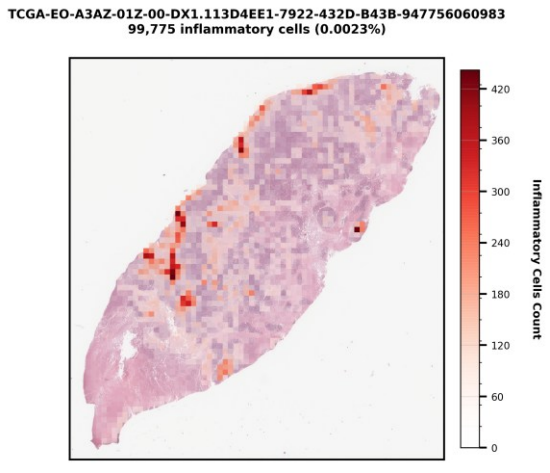
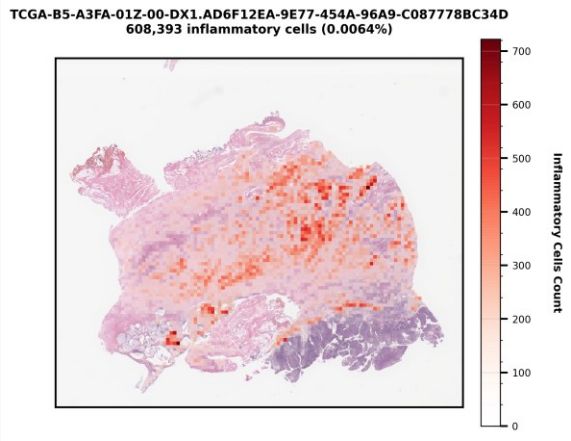
BACKUP

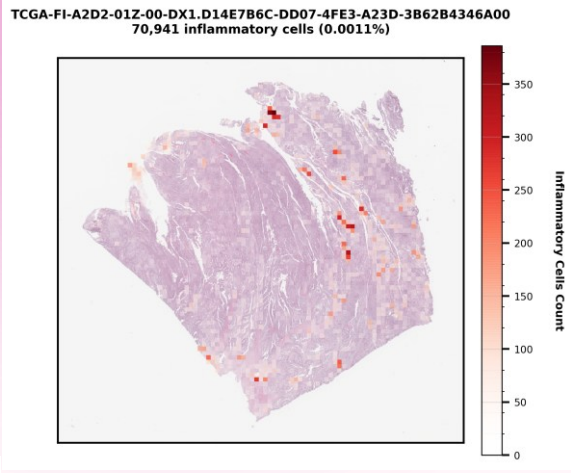
Spatial Heatmaps



MSI-H

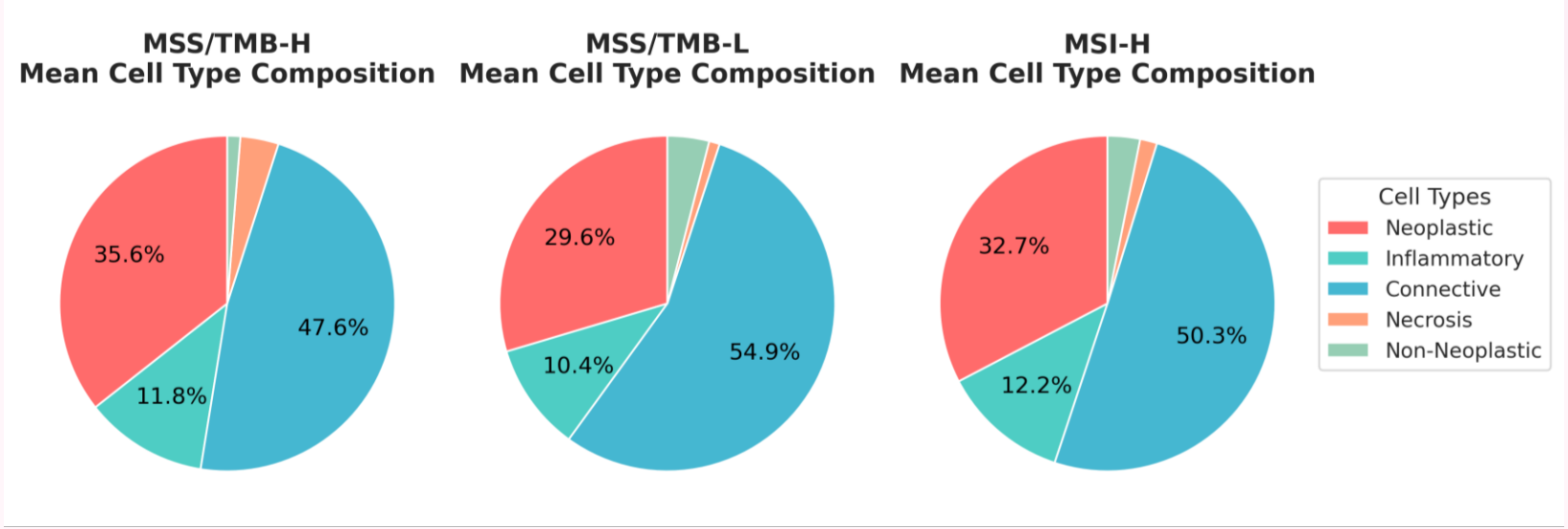
MSS/TMB-H



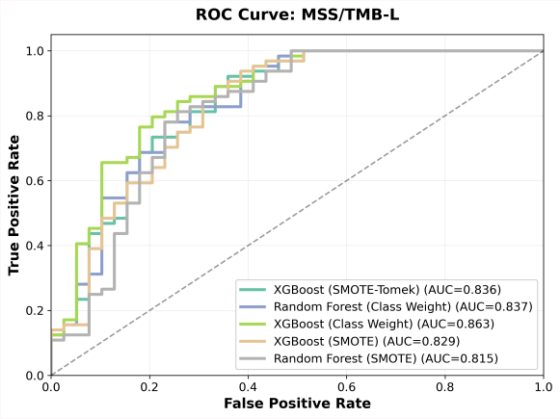


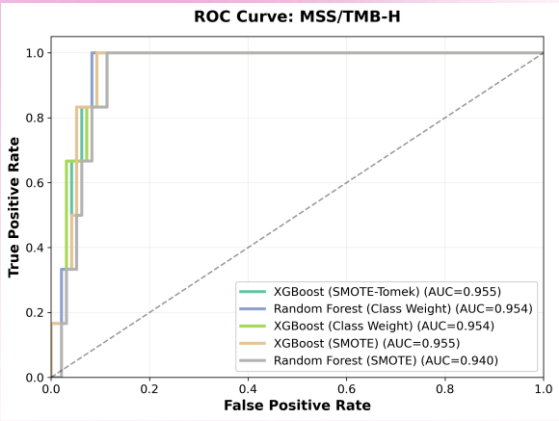
MSS/TMB-L

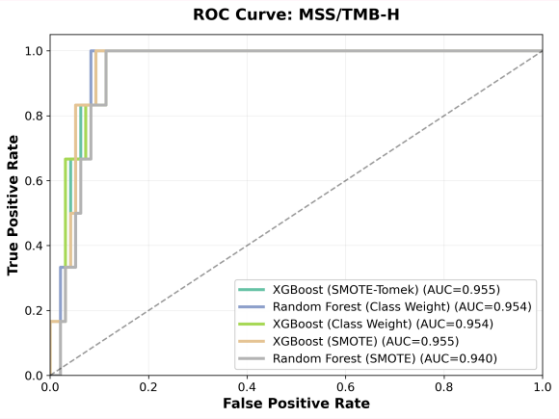
Cellular Composition



ROC-AUC


ROC Curve: MSS/TMB-L
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
XGBoost (SMOTE-Tomek) (AUC=0.836)
Random Forest (Class Weight) (AUC=0.837)
XGBoost (Class Weight) (AUC=0.863)
XGBoost (SMOTE) (AUC=0.829)
Random Forest (SMOTE) (AUC=0.815)


ROC Curve: MSS/TMB-H
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
XGBoost (SMOTE-Tomek) (AUC=0.955)
Random Forest (Class Weight) (AUC=0.954)
XGBoost (Class Weight) (AUC=0.954)
XGBoost (SMOTE) (AUC=0.955)
Random Forest (SMOTE) (AUC=0.940)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XGBoost (SMOTE) (AUC=0.955)
Random Forest (SMOTE) (AUC=0.940)